

Scene Setup

Two scenes are required. Create them via **File → New Scene** and save into `Assets/Scenes/` with the exact names below, then add both to **File → Build Settings → Scenes in Build** (MainMenu = 0, Game = 1).

Scenes are saved by Unity as generated YAML, so they must be authored in the editor. The `.unity.meta` files in this folder reserve stable asset GUIDs; Unity creates/links the matching `.unity` files on first save.

MainMenu.unity

GameObject	Components	Notes
Main Camera	Camera (Orthographic, size 5), AudioListener	Background = solid black
Managers	— (empty parent)	
Managers/GameManager	GameManager	<code>persistAcrossScenes</code> auto-set
Managers/ScoreManager	ScoreManager	persists
Managers/AudioManager	AudioManager	assign clips (see <code>AudioSetup.md</code>)
Managers/SceneLoader	SceneLoader	assign <code>fadeGroup</code> = <code>FadeOverlay CanvasGroup</code>
Background	2-3 × <code>ParallaxBackground</code> children	scroll speeds 0.5 / 1.0 / 2.0
Canvas	Canvas (Screen Space - Overlay), <code>CanvasScaler</code> , <code>GraphicRaycaster</code>	
Canvas/Title	Text (<code>RectTransform</code>)	assign to <code>MainMenuController.titleTransform</code>
Canvas/PlayButton	Button + Text	→ <code>MainMenuController.playButton</code>
Canvas/QuitButton	Button + Text	→ <code>MainMenuController.quitButton</code>
Canvas/HighScoreText	Text	→ <code>MainMenuController.highScoreText</code>
Canvas/FadeOverlay	Image (black, stretch full) + <code>CanvasGroup</code>	alpha 0; → <code>SceneLoader.fadeGroup</code>
MainMenuController	MainMenuController	wire the fields above
EventSystem	EventSystem + <code>StandaloneInputModule</code>	required for UI buttons

Game.unity

GameObject	Components	Notes
Main Camera	Camera (Orthographic, size 5), AudioListener, tag Main-Camera	black background
Managers/GameManager	GameManager	(may persist from MainMenu)
Managers/ScoreManager	ScoreManager	
Managers/AudioManager	AudioManager	
Managers/SceneLoader	SceneLoader	
Managers/BulletPool	BulletPool	assign playerBulletPrefab , enemyBulletPrefab , size 30
Managers/EnemySpawner	EnemySpawner	assign Drone/Fighter/Boss prefabs
Managers/WaveManager	WaveManager	assign spawner = EnemySpawner
Background	2-3 × ParallaxBackground	
Player	Player prefab	tag "Player"
Canvas	Canvas + CanvasScaler + GraphicRaycaster	
Canvas/HUD	HUDController	assign hearts[3], score, wave, multiplier, shield Slider, boss Slider
Canvas/PauseMenu	PauseMenuController	panel + Resume/Restart/MainMenu + 2 volume Sliders
Canvas/GameOver	GameOverController	panel + final/high score Texts + NEW HIGH label + Retry/MainMenu
EventSystem	EventSystem + StandaloneInputModule	

Start-up flow

1. MainMenuController.OnPlay → SceneLoader.LoadScene() + GameManager.NewGame().

2. `GameManager.NewGame()` resets the score, sets state **Playing**, starts `WaveManager.StartWaves()` and plays `bgMusic`.
3. Player death → `PlayerHealth.Die()` → `GameManager.GameOver()` → the `GameOverController` panel appears.
4. **Escape** toggles pause via `GameManager` (freezes `Time.timeScale`).